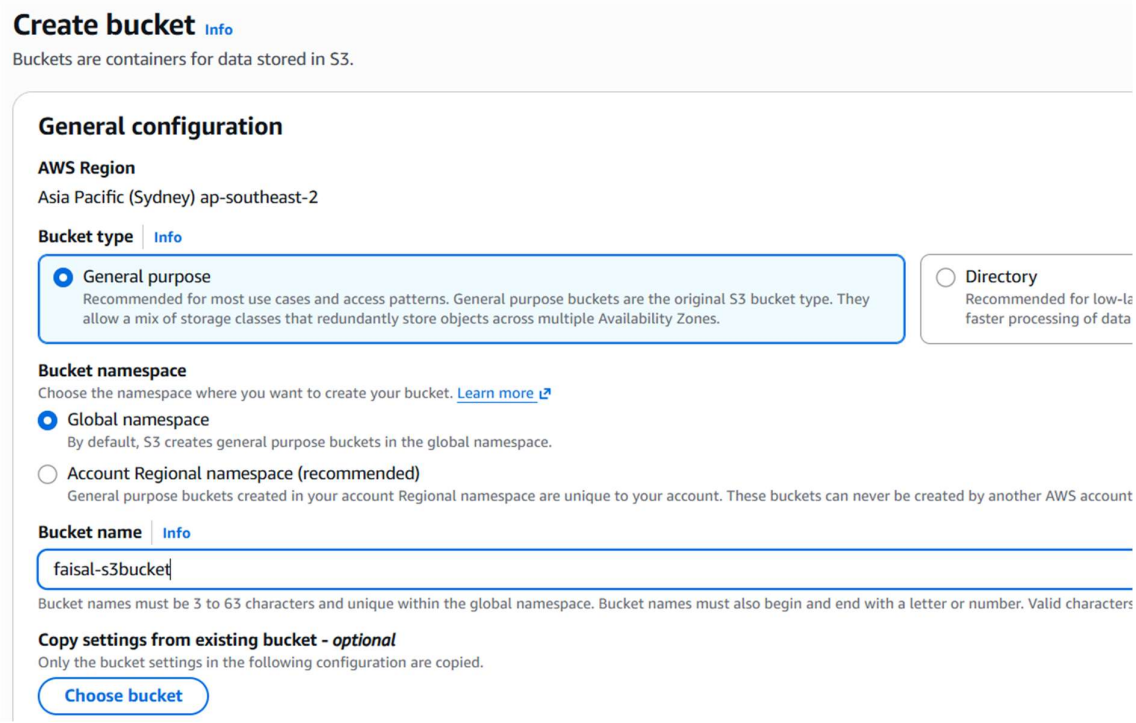
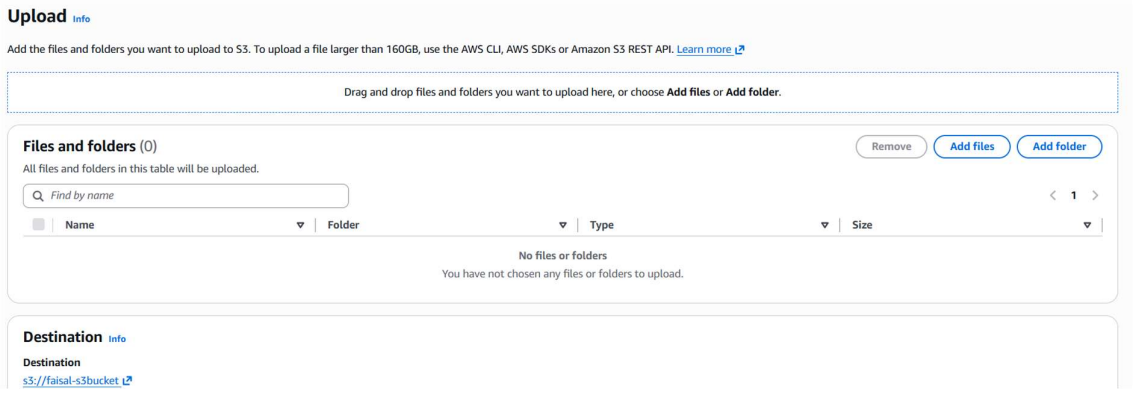
S3 Task

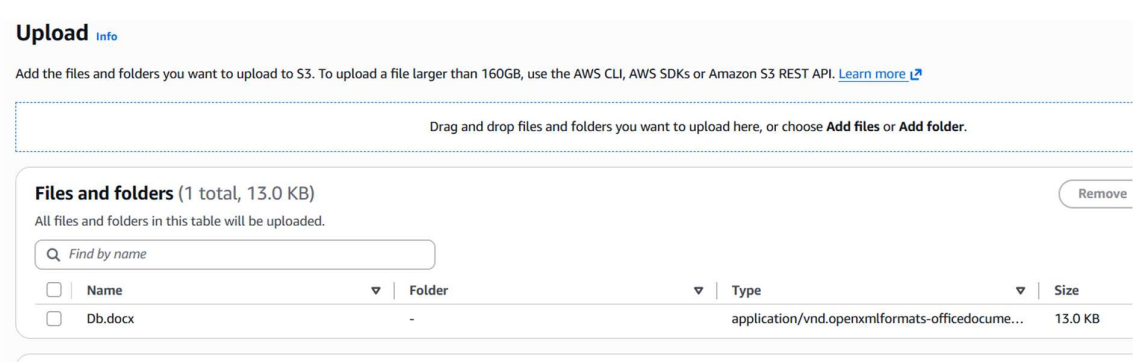
1. Create an S3 bucket and upload some objects to S3.

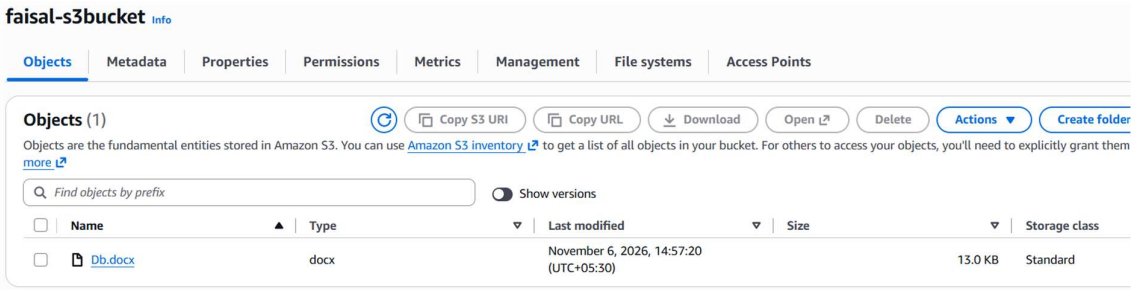
Step 1:- Create a S3 bucket with unique name

Step 2:- Now click upload files. Then add the files, click upload then object will be created.









2. Deploy a static website in the S3 bucket.

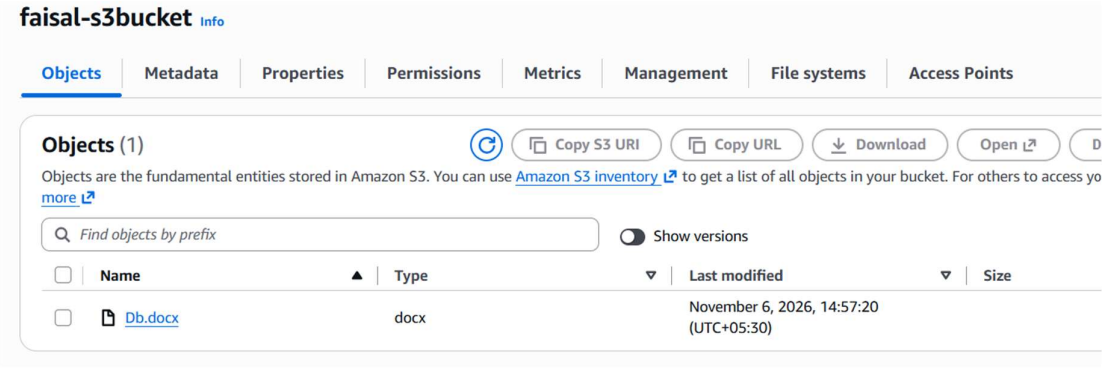
Step 1:- First go to the s3 bucket and upload the Html, JS, CSS files directly into it.

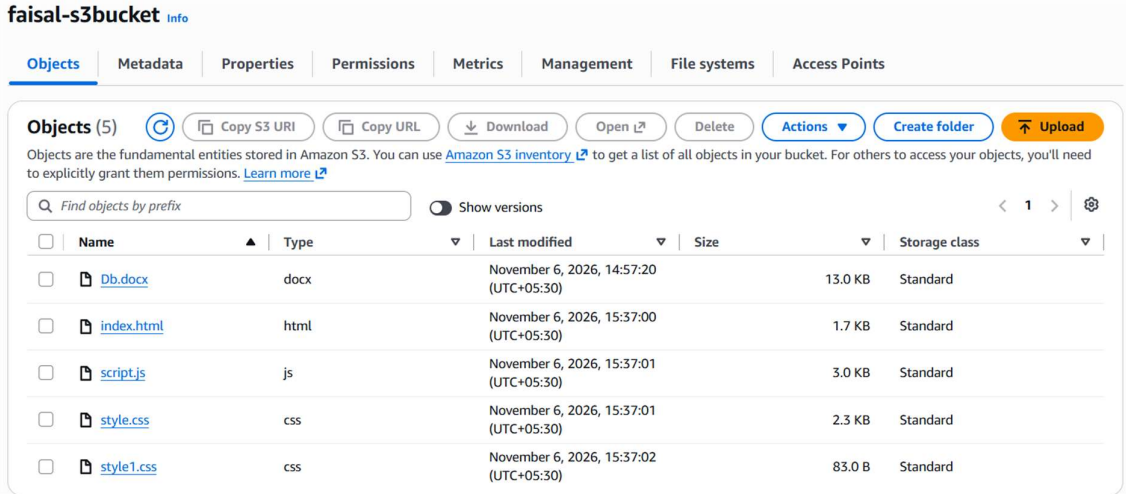
Step 2:- Then go to Properties section scroll to bottom and then enable Static Webhosting.

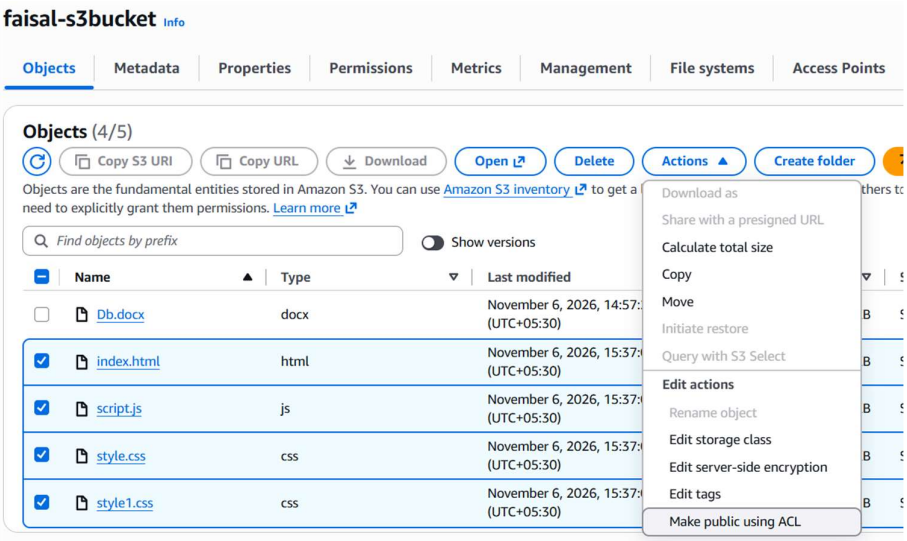
Step 3:- Then just include the name of Html file in the bucket and save.

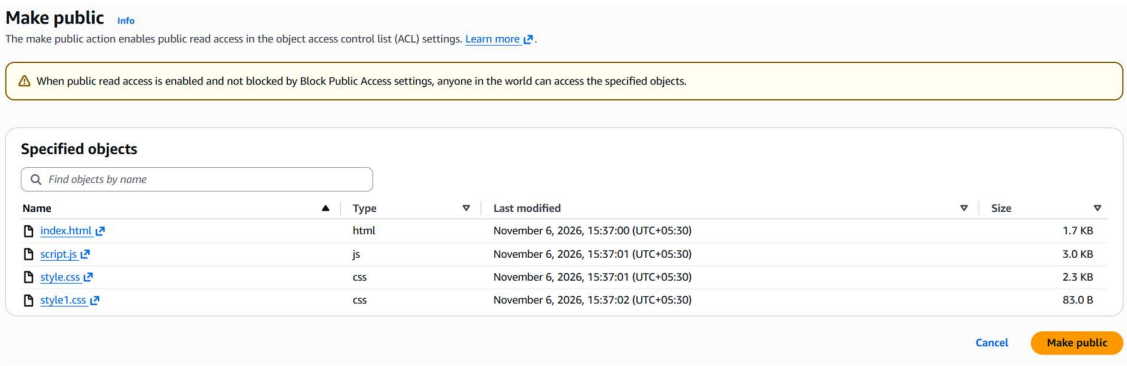
Step 4:- Now go to file and select all the website file and enable public access to them.

Step 5:- Now go to properties and scroll to bottom we can see the website link. Then we can access our website









Disabled

Requester pays — Edit

When enabled, the requester pays for requests and data transfer costs, and anonymous access to this bucket is disabled. Learn more

Requester pays
Disabled

Static website hosting — Edit

Use this bucket to host a website or redirect requests. Learn more

We recommend using AWS Amplify Hosting for static website hosting — Create Amplify app
Deploy a fast, secure, and reliable website quickly with AWS Amplify Hosting. Learn more about Amplify Hosting or View your existing Amplify apps

S3 static website hosting
Disabled

Static website hosting

Use this bucket to host a website or redirect requests. Learn more

Static website hosting

- ○ Disable
- ◉ Enable

Hosting type

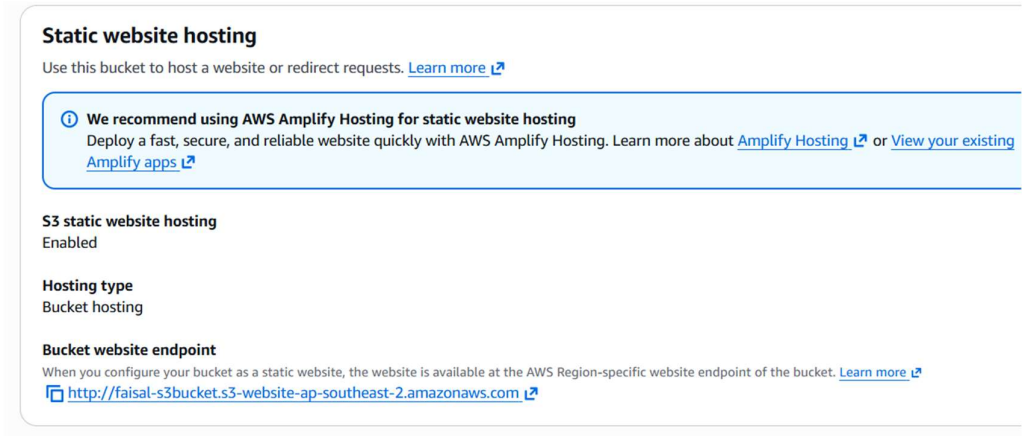
- ◉ Host a static website
  Use the bucket endpoint as the web address. Learn more
- ○ Redirect requests for an object
  Redirect requests to another bucket or domain. Learn more

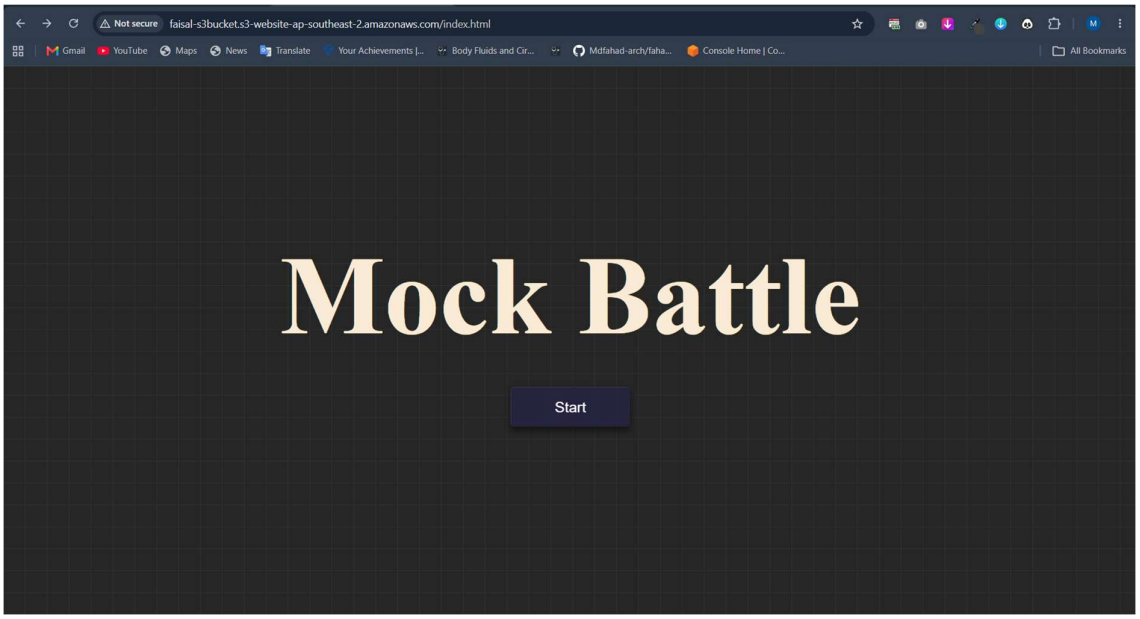
For your customers to access content at the website endpoint, you must make all your content publicly readable. To dc
settings for the bucket. For more information, see Using Amazon S3 Block Public Access

Index document
Specify the home or default page of the website.
index.html

Error document - *optional*
This is returned when an error occurs.
error.html

Redirection rules – *optional*
Redirection rules, written in JSON, automatically redirect webpage requests for specific content. Learn more





3. Enable cross-region replication on S3 buckets.

Step 1:- First create a bucket in two different regions. While creating S3 bucket enable versioning.

Step 2:- Then go to source bucket, then create replication rule in management section.

Step 3:- Name the replication rule, enable the status, select the destination bucket, finally create a new IAM Role.

Step 4:- Select whole bucket to replicate. Now the Replication is done, any file uploaded in source is copied in destination automatically.

Create bucket Info

Buckets are containers for data stored in S3.

General configuration

AWS Region

Asia Pacific (Sydney) ap-southeast-2

Bucket type Info

- (•) General purpose
  Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.
- ( ) Directory
  Recommended for low-lat faster processing of data w

Bucket namespace

Choose the namespace where you want to create your bucket. Learn more

- (•) Global namespace
  By default, S3 creates general purpose buckets in the global namespace.
- ( ) Account Regional namespace (recommended)
  General purpose buckets created in your account Regional namespace are unique to your account. These buckets can never be created by another AWS account.

Bucket name Info

faisal-s3bucket

Bucket names must be 3 to 63 characters and unique within the global namespace. Bucket names must also begin and end with a letter or number. Valid characters a

Copy settings from existing bucket - *optional*

Only the bucket settings in the following configuration are copied.

Choose bucket

Format: s3://bucket/prefix

Object Ownership Info

Control ownership of objects written to this bucket from other AWS accounts and the use of access control lists (ACLs). Object ownership determines who

Object Ownership

- ( ) ACLs disabled (recommended)
  All objects in this bucket are owned by this account. Access to this bucket and its objects is specified using only policies.
- (•) ACLs enabled
  Objects in this bucket can be owned by other AWS accounts. Access to this bucket and its objects can be specified using ACLs.

⚠ We recommend disabling ACLs, unless you need to control access for each object individually or to have the object writer own the data they upload. outside of your account simplifies permissions management and auditing.

Object Ownership

- (•) Bucket owner preferred
  If new objects written to this bucket specify the bucket-owner-full-control canned ACL, they are owned by the bucket owner. Otherwise, they are owned by the object writer.
- ( ) Object writer
  The object writer remains the object owner.

ⓘ If you want to enforce object ownership for new objects only, your bucket policy must specify that the bucket-owner-full-control canned ACL is requ

Block Public Access settings for this bucket

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to t access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these setti without public access. If you require some level of public access to this bucket or objects within, you can customize the individual settings below to suit your specific st

- [ ] **Block *all* public access**
  Turning this setting on is the same as turning on all four settings below. Each of the following settings are independent of one another.
  - [ ] **Block public access to buckets and objects granted through *new* access control lists (ACLs)**
    S3 will block public access permissions applied to newly added buckets or objects, and prevent the creation of new public access ACLs for existing buckets and objects. This setting doe resources using ACLs.
  - [ ] **Block public access to buckets and objects granted through *any* access control lists (ACLs)**
    S3 will ignore all ACLs that grant public access to buckets and objects.
  - [ ] **Block public access to buckets and objects granted through *new* public bucket or access point policies**
    S3 will block new bucket and access point policies that grant public access to buckets and objects. This setting doesn't change any existing policies that allow public access to S3 resou
  - [ ] **Block public and cross-account access to buckets and objects through *any* public bucket or access point policies**
    S3 will ignore public and cross-account access for buckets or access points with policies that grant public access to buckets and objects.

⚠ **Turning off block all public access might result in this bucket and the objects within becoming public**

AWS recommends that you turn on block all public access, unless public access is required for specific and verified use cases such as static website hosting.

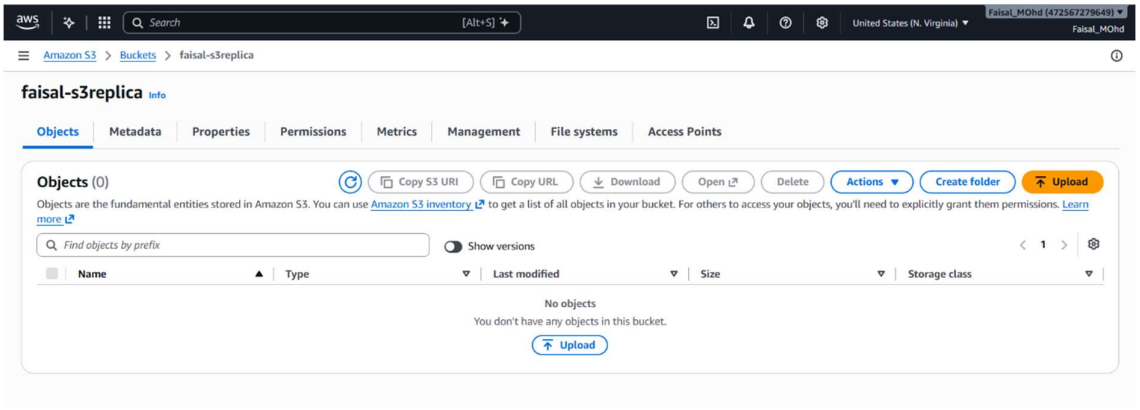
- [x] I acknowledge that the current settings might result in this bucket and the objects within becoming public.

Bucket Versioning

Versioning is a means of keeping multiple variants of an object in the same bucket
can easily recover from both unintended user actions and application failures. Lea

Bucket Versioning

- ○ Disable
- ◉ Enable



Create replication rule Info

Use replication rules to define options you want Amazon S3 to apply during replication such as server-side encryption, replica ownership, transitioni
annotations attached to your objects. Learn more

ⓘ Replication now includes replicating any annotations attached to your objects, which might increase your costs. To understand the potential c

Replication rule configuration

Replication rule name

s3-replica

Up to 255 characters. In order to be able to use CloudWatch metrics to monitor the progress of your replication rule, the replication rule name must only contain Englis

Status

Choose whether the rule will be enabled or disabled when created.

- ◉ Enabled
- ○ Disabled

Priority

The priority value resolves conflicts that occur when an object is eligible for replication under multiple rules to the same destination. The rule is added to the configurat

0

Destination

Destination

You can replicate objects across buckets in different AWS Regions (Cross-Region Replication) or you can replicate objects across buckets in the sam
configuration. Learn more or see Amazon S3 pricing

- ◉ Choose a bucket in this account
- ○ Specify a bucket in another account

Bucket name

s3://faisal-s3replica

Choose the bucket that will receive replicated objects.

Destination Region

US East (N. Virginia) us-east-1

IAM role

The selected IAM role applies to all rules in the bucket.

Permission to access the specified resources

- (●) Create new role
- ( ) Choose from existing IAM roles
- ( ) Enter IAM role ARN

Create Batch Operations job

Job settings

A job is used to execute batch operations on a list of S3 objects. The list of objects is contained in a replication mani

Job run options

You can choose whether to have the job start automatically after the replication manifest is generated or to have the job wait in the *A*

- (●) Automatically run the job when it's ready
  When selected, the job automatically runs without waiting for you to start it.

Completion report

Generate a CSV completion report that lists your target objects, task success or error codes, outputs, and descriptior

- [x] Generate completion report

Completion report scope

- ( ) Failed tasks only
- (●) All tasks

Completion report destination account

- (●) This account
- ( ) A different account

Destination

Enter a destination in Amazon S3 where your object(s) are stored. Amazon S3 is object storage built to store and retrieve any amount

Destination

Enter a destination in Amazon S3 where your object(s) are stored. Amazon S3 is object storage built to store and retrieve any amount of data from anywhere.

s3://faisal-s3replica | View | Browse S3

Format: s3://<bucket>/<optional-prefix-with-path>. S3 will append the path with a "/". If you add a "/" to the prefix, it will appear as an extra folder in the S3 console.

Bucket owner account ID

472567279649

Completion report destination

s3://faisal-s3replica/job-<job-id>/results/<unique-id>.csv

Permissions

Choose an IAM role with the required access permissions and trust relationships . An IAM role policy template based on your job configuration, and the IAM trust policy required for batch operations to assume the IAM role are available below. Learn more about IAM roles.

▶ **View IAM role policy template and IAM trust policy**

Permission to access the specified resources

- (●) Create new role
- ( ) Choose from existing IAM roles
- ( ) Enter IAM role ARN

Cancel | Save

Amazon S3 > Buckets > faisal-s3replica

faisal-s3replica Info

Objects | Metadata | Properties | Permissions | Metrics | Management | File systems | Access Points

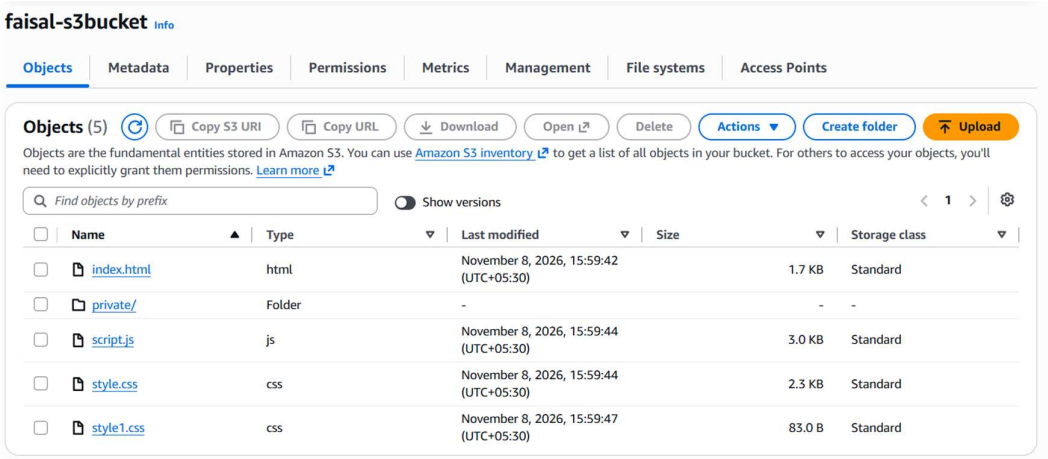
Objects (6)

Copy S3 URI | Copy URL | Download | Open | Delete | Actions | Create folder

Objects are the fundamental entities stored in Amazon S3. You can use Amazon S3 inventory to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them pe more

Find objects by prefix | Show versions

Name	Type	Last modified	Size	Storage class
index.html	html	November 8, 2026, 15:59:42 (UTC+05:30)	1.7 KB	Standard
job-8339ecae-9160-468d-bdb1-bf9fdbc5d493/	Folder	-	-	-
private/	Folder	-	-	-
script.js	js	November 8, 2026, 15:59:44 (UTC+05:30)	3.0 KB	Standard
style.css	css	November 8, 2026, 15:59:44 (UTC+05:30)	2.3 KB	Standard
style1.css	css	November 8, 2026, 15:59:47 (UTC+05:30)	83.0 B	Standard

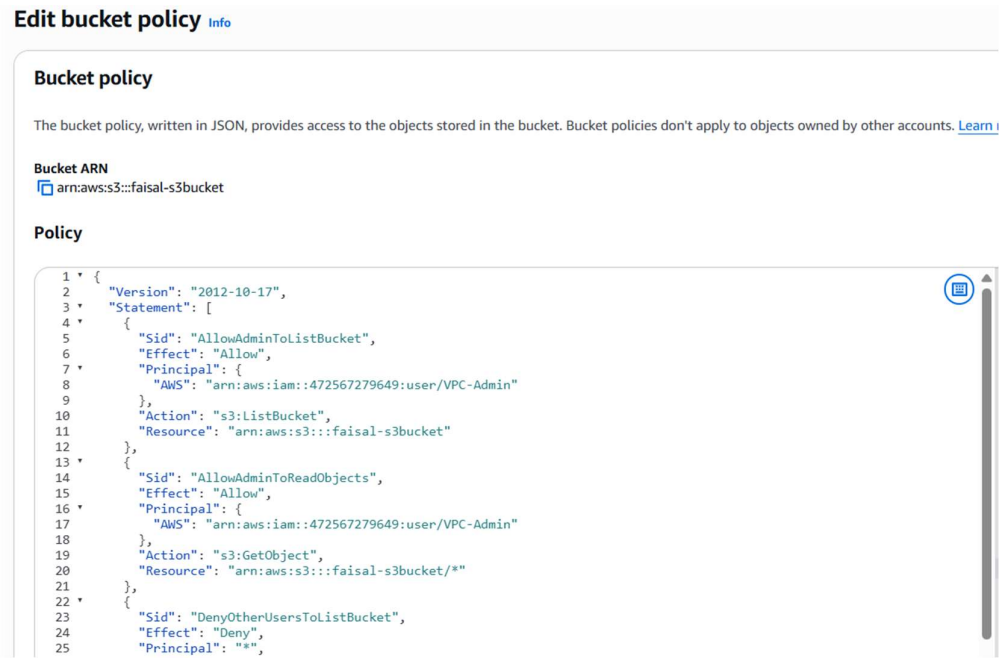


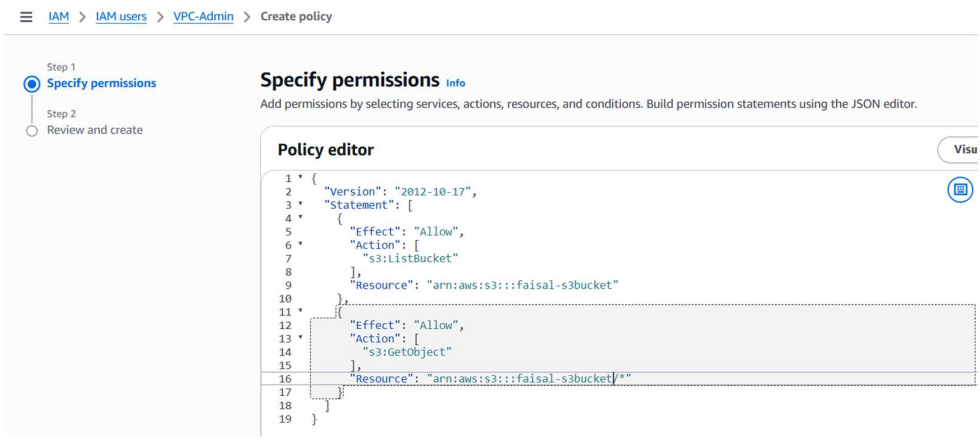
4. Configure a bucket policy so only the Admin user can see the objects of the S3 bucket.

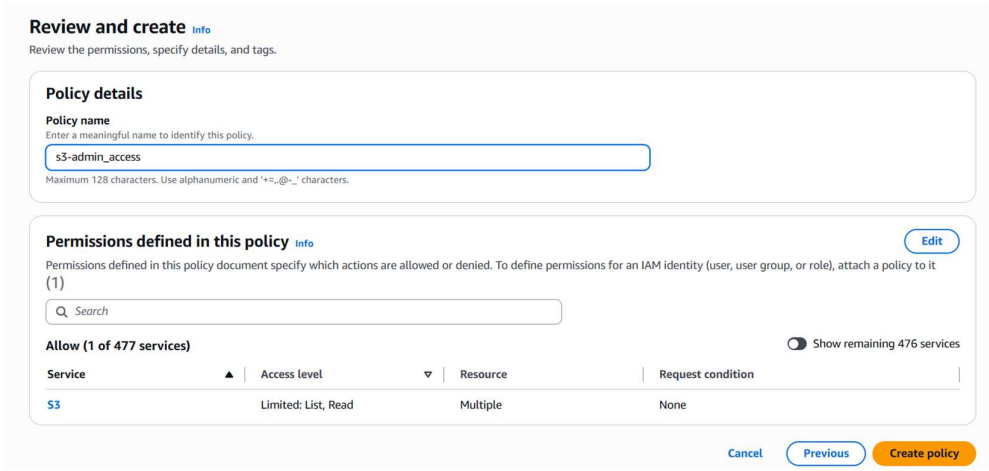
Step 1:- Select a S3 bucket and create a bucket policy where only Admin user has all access to S3 bucket.

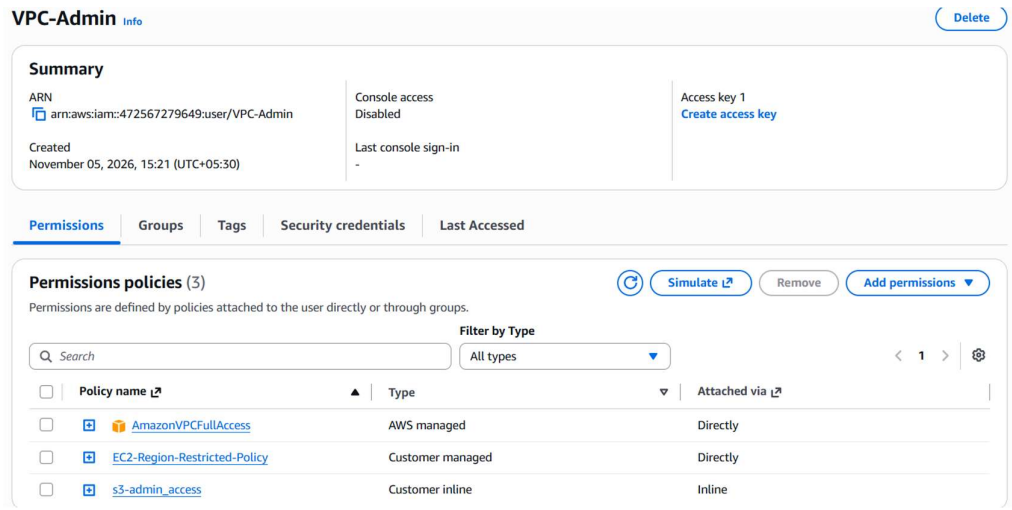
Step 2:- Then go to IAM user and create inline policy where the user request the access of S3 bucket.

Step 3:- Now try to access the s3 bucket with other user to check the policy was added successfully.










```
Faisal@LAPTOP-4DO8SS4E UCRT64 ~
$ aws sts get-caller-identity
{
    "UserId": "AIDAW4BZUOAQT5SUQWXLI",
    "Account": "472567279649",
    "Arn": "arn:aws:iam::472567279649:user/Faisal"
}


Faisal@LAPTOP-4DO8SS4E UCRT64 ~
$ aws s3 ls s3://faisal-s3bucket

aws: [ERROR]: An error occurred (AccessDenied) when calling the ListObjectsV2 operation: User: arn:aws:iam::4725
67279649:user/Faisal is not authorized to perform: s3:ListBucket on resource: "arn:aws:s3:::faisal-s3bucket" wit
h an explicit deny in a resource-based policy
```

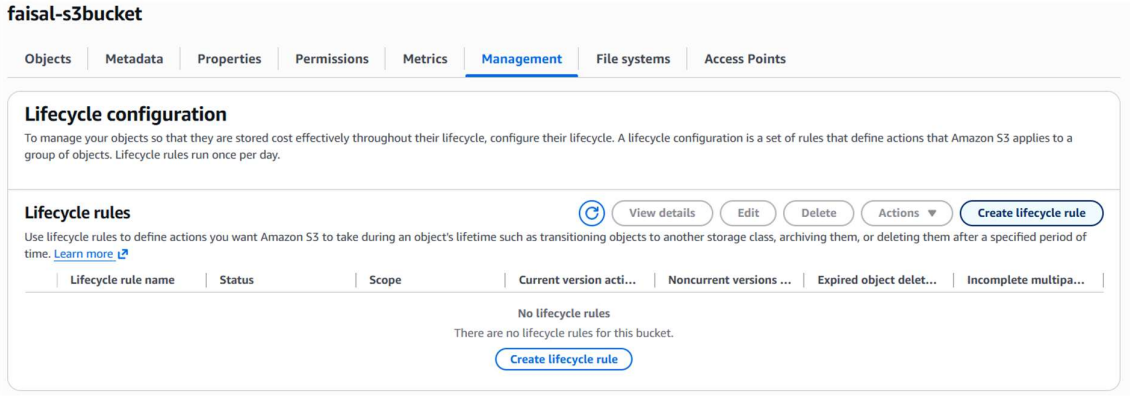
5. Set up lifecycle policies to automatically transition or delete objects based on specific criteria.

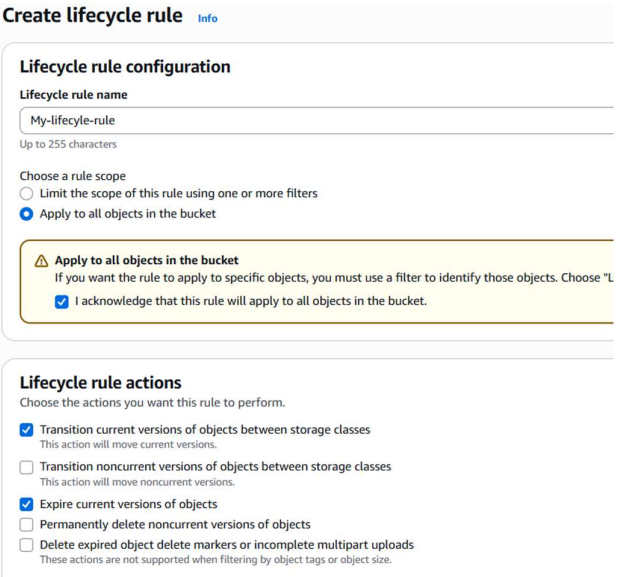
Step 1:- Go to management service of S3 bucket then click on create Lifecycle policy.

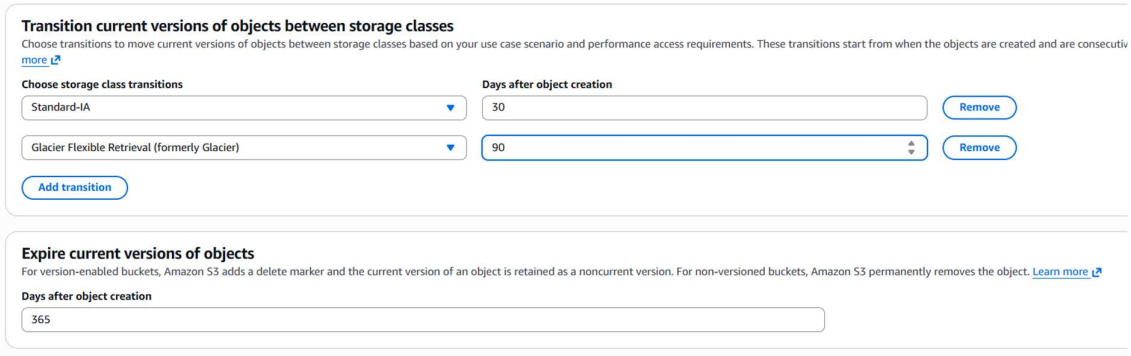
Step 2:- name the policy, enable apply to all objects in bucket. Then go to transition rule actions enable transition current version of objects between and Expire current versions of objects.

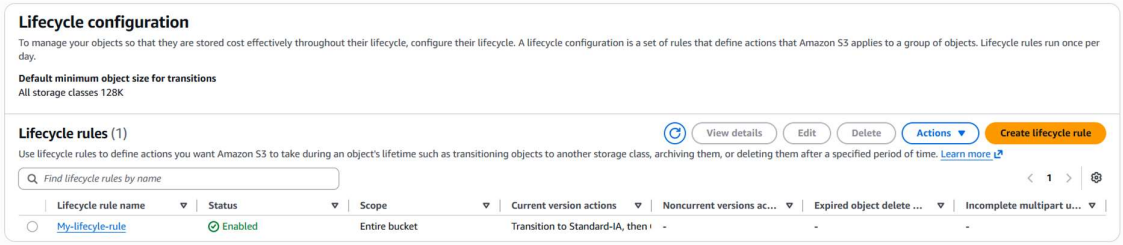
Step 3:- Then create transition rules according time period. Add expire rule time period.

Step 4:-Then our Lifecycle policy is created.









6. Push some objects to S3 using the AWS CLI.

Step 1:- check if the configured user has s3 bucket access or not.

Step 2:- select the file and use aws s3 cp file-name s3bucket_route in cli

Step 3:- check whether the file was copied successfully or not.

7. Write a Bash script to create an S3 bucket.

Step 1:- connect the IAM user which has S3 access.

Step 2:- Write a bash script to create S3 bucket. First take input for bucket name, then the Region input.

Step 3:- Then validate if the S3 bucket exists or not, and then check the name is in correct name or not.

Step 4:- Then run the script and create S3 bucket. Then go to console and check if it was created or not.

```
Faisal@LAPTOP-4DO8SS4E UCRT64 ~/Downloads/AWS-scripts
$ aws sts get-caller-identity
{
    "UserId": "AIDAW4BZUOAQT5SUQWXLI",
    "Account": "472567279649",
    "Arn": "arn:aws:iam::472567279649:user/Faisal"
}
```

```
Faisal@LAPTOP-4DO8SS4E UCRT64 ~/Downloads/AWS-scripts
$ vi s3bucket-create.bash

Faisal@LAPTOP-4DO8SS4E UCRT64 ~/Downloads/AWS-scripts
$ cat s3bucket-create.bash
#!/bin/bash

echo "Starting the Process"

# Step 1: Check whether AWS CLI is installed
if ! which aws &> /dev/null
then
    echo "AWS CLI is not installed!"
    exit 1
fi

echo "AWS CLI is installed."

# Step 2: Check AWS credentials
if ! aws sts get-caller-identity &> /dev/null
then
    echo "AWS credentials are not configured or are invalid!"
    exit 1
fi

echo "AWS credentials verified."

# Step 3: Get bucket name from user
read -p "Enter S3 bucket name: " bucket_name

# Step 4: Convert uppercase letters to lowercase
bucket_name=$(echo "$bucket_name" | tr '[:upper:]' '[:lower:]')

# Step 5: Validate bucket name
if [[ ! "$bucket_name" =~ ^[a-z0-9][a-z0-9.-]{1,61}[a-z0-9]$ ]]
then
    echo "Invalid bucket name!"
    echo "Bucket name must be 3-63 characters and use lowercase letters, numbers, dots or hyphens."
    exit 1
fi

# Additional S3 naming restrictions
if [[ "$bucket_name" == *".."* || "$bucket_name" =~ ^[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+$ ]]
then
    echo "Invalid S3 bucket name!"
```

```
# Additional S3 naming restrictions
if [[ "$bucket_name" == *".."* || "$bucket_name" =~ ^[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+$ ]]
then
    echo "Invalid S3 bucket name!"
    exit 1
fi

echo "Bucket name: $bucket_name"

# Step 6: Check whether bucket already exists
if aws s3api head-bucket --bucket "$bucket_name" 2>/dev/null
then
    echo "Bucket already exists, or you already have accesss to it."
    exit 0
fi

# Step 7: Get AWS region
read -p "Enter AWS region (default: ap-south-1): " region
region=${region:-ap-south-1}

# Step 8: Create bucket
if [[ "$region" == "us-east-1" ]]
then
    aws s3api create-bucket \
        --bucket "$bucket_name" \
        --region "$region"
else
    aws s3api create-bucket \
        --bucket "$bucket_name" \
        --region "$region" \
        --create-bucket-configuration LocationConstraint="$region"
fi

# Step 9: Check creation status
if [ $? -eq 0 ]
then
    echo "S3 bucket created successfully: $bucket_name"
else
    echo "Failed to create S3 bucket!"
fi
```

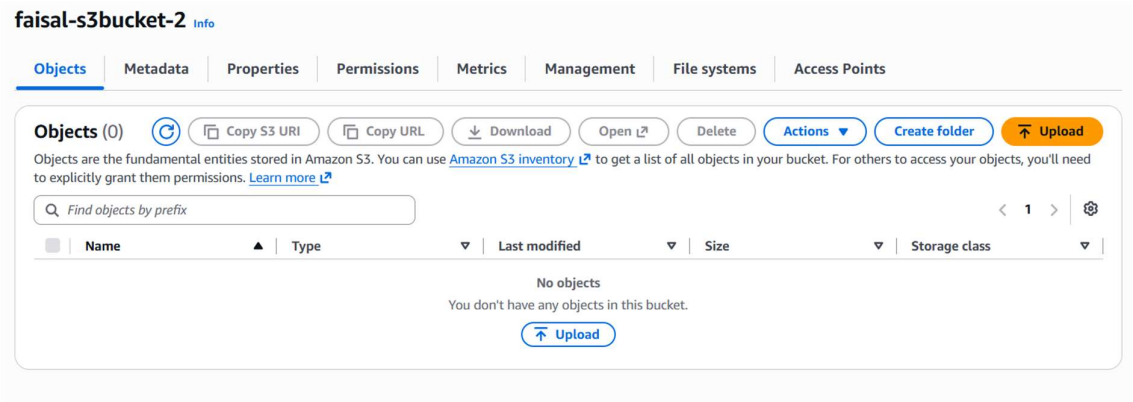
```
Faisal@LAPTOP-4DO8SS4E UCRT64 ~/Downloads/AWS-scripts
$ chmod +x s3bucket-create.bash

Faisal@LAPTOP-4DO8SS4E UCRT64 ~/Downloads/AWS-scripts
$ ./s3bucket-create.bash
Starting the Process
AWS CLI is installed.
AWS credentials verified.
Enter S3 bucket name: faisal-2
Bucket name: faisal-2
Enter AWS region (default: ap-south-1):

aws: [ERROR]: An error occurred (BucketAlreadyExists) when callin
e is shared by all users of the system. Please select a different
Failed to create S3 bucket!

Faisal@LAPTOP-4DO8SS4E UCRT64 ~/Downloads/AWS-scripts
$ ./s3bucket-create.bash
Starting the Process
AWS CLI is installed.
AWS credentials verified.
Enter S3 bucket name: faisal-s3bucket-2
Bucket name: faisal-s3bucket-2
Enter AWS region (default: ap-south-1):
{
    "Location": "http://faisal-s3bucket-2.s3.amazonaws.com/",
    "BucketArn": "arn:aws:s3:::faisal-s3bucket-2"
}

S3 bucket created successfully: faisal-s3bucket-2
```



8. Upload a 1 GB file to S3 using the CLI.

Step 1:- First check if the IAM user configured has upload access or not.

Step 2:- Then select the 1gb file and upload the file with CLI

Step 3:- After some time the file is uploaded. Go to console and check the file in S3 bucket.

```
Faisal@LAPTOP-4DO8SS4E UCRT64 /d
$ ls
'$AV_ASW'/                                          Error.png                 'Study/'
'$RECYCLE.BIN'/                                    'Gen AI'/                 'System Volume Information'/
'ALLEN ALL TESTS 23-24-20260924T173101Z-1-001'/     Git/                      'Task/'
'ALLEN MAJOR ALL CLASSROOM-20260924T173706Z-1-001'/ 'Kicad Libraries.pretty'/ 'Task 1.docx'
 Biology-20260924T173408Z-1-001/                    MapData/                  Task.zip
 Blender/                                           MobaXterm_Portable_v22.1/ 'Tasks/'
'Da vinci'/                                        'Program Files'/          'Topper Notes-20260924T174040Z-1-001'/

Faisal@LAPTOP-4DO8SS4E UCRT64 /d
$ aws sts get-caller-identity
{
    "UserId": "AIDAW4BZUOAQT5SUQWXLI",
    "Account": "472567279649",
    "Arn": "arn:aws:iam::472567279649:user/Faisal"
}


Faisal@LAPTOP-4DO8SS4E UCRT64 /d
$ aws s3 cp Task.zip s3://faisal-s3bucket/
upload: .\Task.zip to s3://faisal-s3bucket/Task.zip
```

